

# Florian De Rop

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## EXPERIENCE

### Technical Staff

Retro Biosciences Oct. 2023 – Present  
*Organoid culture automation, bioreactors, single-cell sequencing, bioinformatics*

### PhD Researcher

KU Leuven Oct. 2019 – Sep. 2023  
*Design, development and benchmarking of single-cell sequencing assays*

### Master Thesis Tutor

KU Leuven Sep. 2017 – 2020

### Production Assistant

IDT Sep. 2017, Jul. 2018  
*Manufacturing oligonucleotides*

### Research Intern

KU Leuven Stem Cell Institute Sep. 2015  
*HuH-7 gene expression response to various osmolytes*

## EDUCATION

### MSc. Bioscience Engineering

KU Leuven – Magna cum laude 2017 – 2019  
*Optimisation of inDrop single-cell RNA-seq and development of a single-cell ATAC-seq protocol*

### KU Leuven Honours Programme

Institute for the Future 2018 – 2019  
*Vaccination dynamics*

### BSc. Biomedical Sciences

KU Leuven 2013 – 2016

## PUBLICATIONS

- 2023 **Benchmarking of single-cell ATAC sequencing tools**  
*De Rop & Heyn, Nature Biotechnology*
- 2023 **Systematic benchmarking of single-cell ATAC-sequencing protocols**  
*De Rop et al., Nature Biotechnology*
- 2022 **Decoding gene regulation in the fly brain**  
*Janssens et al., Nature*
- 2021 **HyDrop: droplet-based scATAC-seq and scRNA-seq using dissolvable hydrogel beads**  
*De Rop et al., eLife*
- 2018 **Creasensor: SIMPLE technology for creatinine detection in plasma.**  
*Dal Dosso et al., Analytica Chimica Acta*

## AS A SPEAKER

- 2023 **Revolutionising Next Generation Sequencing**
- 2023 **AGBT**
- 2022 **Inauguration of the Leuven Institute for Single-Cell Omics**
- 2022 **Single Cell Genomics Utrecht**
- 2021 **CRG visiting researcher**
- 2021 **EPFL visiting researcher**
- 2021 **MJFF ASAP COSA**
- 2021 **LifeTime conference**

## AWARDS

- 2021 **Aligning Science Across Parkinson's**  
*Open Science Champion*
- 2020 **VIB Tech Watch**  
*Technology Development Grants I, II, III*
- 2019 **Research Foundation Flanders**  
*Strategic Research Grant*
- 2018 **SensUs Biosensor Competition**  
*1st Prize in Translation Potential*
- 2016 **SensUs Biosensor Competition**  
*1st Prize in Analytical Performance*  
*1st Prize in Translation Potential*  
*1st Prize in Public Inspiration*

## SOFTWARE & LAB PROTOCOLS

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|                     |                                                                  |
|---------------------|------------------------------------------------------------------|
| <b>PUMATAC</b>      | NextFlow pipeline for universal alignment of scATAC-seq data     |
| <b>scforest</b>     | Overview of scTech                                               |
| <b>HyDrop-ATAC</b>  | Droplet-based scATAC-seq                                         |
| <b>HyDrop-RNA</b>   | Droplet-based scRNA-seq                                          |
| <b>HyDrop Beads</b> | Dissolvable polyacrylamide hydrogel bead synthesis and barcoding |

## LANGUAGES

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|                |                      |
|----------------|----------------------|
| <b>English</b> | CEFR C2 Professional |
| <b>Dutch</b>   | Native               |
| <b>French</b>  | Basic                |